

Problem Set 5: Answers

Probability and Statistical Inference

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1. Let $Z \sim N(0, 1)$, let $I_n \sim \text{Bernoulli}(1/n^2)$, and let U equal -1 or 1 with equal probability. The following are three sequences of estimators for a fixed target β :

$$A_n = \beta + \frac{Z}{\sqrt{n}}, \quad B_n = \beta + nI_n, \quad C_n = \beta + U.$$

- (a) For each estimator, calculate its bias, variance, and mean squared error.
- (b) For each sequence, determine whether it converges to β in mean square and whether it converges to β in probability. Also state its limiting distribution.
- (c) Find the distribution of $\sqrt{n}(A_n - \beta)$ for every n . Explain what this tells us beyond the statement $A_n \xrightarrow{p} \beta$.
- (d) Which sequence shows that convergence in probability need not imply convergence in mean square? Explain the role of rare, large errors.
- (e) You may use the following probability bound: $\Pr(W > a) \leq \mathbb{E}[W]/a$ for any random variable $W \geq 0$ and any $a > 0$. For an arbitrary sequence T_n , apply it to $(T_n - \beta)^2$ to prove that $T_n \xrightarrow{\text{m.s.}} \beta$ implies $T_n \xrightarrow{p} \beta$.

Solution.

- (a) For A_n ,

$$\text{Bias}(A_n) = 0, \quad \text{Var}(A_n) = \frac{1}{n}, \quad \text{MSE}(A_n) = \frac{1}{n}.$$

For B_n , using $p_n = 1/n^2$,

$$\begin{aligned} \text{Bias}(B_n) &= \mathbb{E}(nI_n) = np_n = \frac{1}{n}, \\ \text{Var}(B_n) &= n^2 p_n (1 - p_n) = 1 - \frac{1}{n^2}, \\ \text{MSE}(B_n) &= \mathbb{E}[(nI_n)^2] = n^2 p_n = 1. \end{aligned}$$

The identity $\text{MSE} = \text{Var} + \text{Bias}^2$ gives the same last result. Finally,

$$\text{Bias}(C_n) = 0, \quad \text{Var}(C_n) = 1, \quad \text{MSE}(C_n) = 1.$$

- (b) Since $\mathbb{E}[(A_n - \beta)^2] = 1/n \rightarrow 0$, $A_n \xrightarrow{\text{m.s.}} \beta$ and hence $A_n \xrightarrow{p} \beta$; it also converges in distribution to the constant β . For any fixed $\varepsilon > 0$ and all $n > \varepsilon$,

$$\Pr(|B_n - \beta| > \varepsilon) = \Pr(I_n = 1) = \frac{1}{n^2} \rightarrow 0.$$

Thus $B_n \xrightarrow{p} \beta$ and $B_n \xrightarrow{d} \beta$, but its MSE is always 1, so it does not converge in mean square. For $0 < \varepsilon < 1$,

$$\Pr(|C_n - \beta| > \varepsilon) = 1,$$

so C_n is not consistent and does not converge in mean square. Its distribution is the same two-point distribution for every n , so $C_n \xrightarrow{d} \beta + U$.

- (c) Exactly, not merely asymptotically,

$$\sqrt{n}(A_n - \beta) = Z \sim N(0, 1).$$

Consistency only says that the unscaled error becomes small. The scaled statement also gives the limiting shape and the root- n rate of the error. This information can be used for approximate standard errors, confidence intervals, and tests.

- (d) The sequence B_n is the counterexample. Its nonzero error occurs with probability only $1/n^2$, which vanishes, but when it occurs the error has size n . Squaring produces n^2 , exactly offsetting its probability: $n^2(1/n^2) = 1$.
- (e) If $T_n \xrightarrow{\text{m.s.}} \beta$, then $\mathbb{E}[(T_n - \beta)^2] \rightarrow 0$. For every $\varepsilon > 0$, the supplied probability bound gives

$$\Pr(|T_n - \beta| > \varepsilon) = \Pr((T_n - \beta)^2 > \varepsilon^2) \leq \frac{\mathbb{E}[(T_n - \beta)^2]}{\varepsilon^2} \rightarrow 0.$$

This is precisely $T_n \xrightarrow{p} \beta$.

2. This question explores why both parts of “independent and identically distributed” matter.

- (a) Suppose $X_i = \mu + A + e_i$. Here A is one mean-zero common shock, with finite variance, that appears in every observation. The e_i are iid with mean zero and finite variance, and A is independent of every e_i . Find the probability limit of $\bar{X}_n$. For $i \neq j$, calculate $\text{Cov}(X_i, X_j)$ and explain why averaging does not remove A .
- (b) Now suppose the X_i are independent and $X_i \sim N(0, i^a)$, where the second argument is the variance and $a \geq 0$. Show that

$$\text{Var}(\bar{X}_n) = \frac{1}{n^2} \sum_{i=1}^n i^a, \quad \frac{\text{Var}(\bar{X}_n)}{n^{a-1}/(a+1)} \rightarrow 1.$$

You may compare the sum with $\int_0^n x^a dx$ and $\int_1^{n+1} x^a dx$. For which values of a does $\bar{X}_n \xrightarrow{p} 0$? What happens when $a = 1$ and when $a > 1$?

- (c) A researcher samples only from one friendship network. The sample average approaches $\mu + g$, but the mean in the target population is μ . Explain separately the problem caused by dependence and the problem caused by the shift g . Must a larger sample from the same network solve either problem?
- (d) State weak stationarity in terms of the mean, variance and covariances at different lags of a time series. Briefly explain why stationarity together with suitable weak dependence can replace iid sampling in a time-series law of large numbers.

Solution.

- (a) The average is

$$\bar{X}_n = \mu + A + \bar{e}_n.$$

The iid law of large numbers gives $\bar{e}_n \xrightarrow{p} 0$, so $\bar{X}_n \xrightarrow{p} \mu + A$, which is a random limit. Unless $A = 0$ with probability one, the average does not converge to μ . For $i \neq j$,

$$\begin{aligned} \text{Cov}(X_i, X_j) &= \text{Cov}(A + e_i, A + e_j) \\ &= \text{Var}(A), \end{aligned}$$

because all other cross-covariances are zero. Every observation contains the same A . Therefore the average contains A itself, not A/n .

(b) Independence gives

$$\text{Var}(\bar{X}_n) = \frac{1}{n^2} \sum_{i=1}^n \text{Var}(X_i) = \frac{1}{n^2} \sum_{i=1}^n i^a.$$

Since x^a is non-decreasing on $[0, \infty)$,

$$\frac{n^{a+1}}{a+1} \leq \sum_{i=1}^n i^a \leq \frac{(n+1)^{a+1} - 1}{a+1}.$$

Dividing by $n^{a+1}/(a+1)$ shows that the ratio tends to one, giving the requested limit for the variance. If $0 \leq a < 1$, the variance tends to zero. Since the mean is zero, this gives mean-square and probability convergence to zero. If $a = 1$,

$$\text{Var}(\bar{X}_n) = \frac{n(n+1)/2}{n^2} \rightarrow \frac{1}{2}.$$

Each average is normal, so $\bar{X}_n \xrightarrow{d} N(0, 1/2)$ rather than concentrating at zero. If $a > 1$, its variance diverges; indeed, normality then implies $\Pr(|\bar{X}_n| \leq \varepsilon) \rightarrow 0$ for every fixed $\varepsilon > 0$. Independence alone has not controlled the changing distributions.

- (c) Strong positive dependence means that observations from additional friends may provide little new information. As a result, the sample average may approach its limit slowly, or may fail to approach a constant if there is a persistent common component. The shift g is a separate target or selection problem: even if the average concentrates perfectly, it does so around the network mean $\mu + g$, not the target-population mean μ . A larger sample from the same network cannot remove g , and it need not remove persistent dependence. A more appropriate sampling design and assumptions about dependence are needed.
- (d) A weakly stationary series has finite second moments and satisfies, for every date t ,

$$\mathbb{E}(X_t) = \mu, \quad \text{Var}(X_t) = \gamma_0, \quad \text{Cov}(X_t, X_{t-h}) = \gamma_h,$$

so the mean and variance do not depend on calendar time, and covariance depends only on the lag. Stationarity makes the first and second moments stable over time. Suitable weak dependence ensures that distant observations provide enough new information for the variance of an average to vanish. Stationarity by itself is not sufficient: a perfectly persistent stationary common shock is a counterexample.

3. Consider the linear model $y_i = x_i' \beta + \varepsilon_i$, where x_i is a $k \times 1$ vector and the data matrix X has full column rank (A1 and A2). Assume the expectations used below exist.

- (a) Starting from $\hat{\beta} = (X'X)^{-1}X'y$, derive the exact decomposition of $\hat{\beta} - \beta$. Under A3, $\mathbb{E}(\varepsilon | X) = 0$, establish finite-sample conditional and unconditional unbiasedness.
- (b) Rewrite the decomposition using sample averages. Suppose laws of large numbers give

$$\frac{1}{n} \sum_{i=1}^n x_i x_i' \xrightarrow{p} Q, \quad \frac{1}{n} \sum_{i=1}^n x_i \varepsilon_i \xrightarrow{p} g,$$

where Q is nonsingular. Find $\text{plim } \hat{\beta}$ and state the condition for consistency. You may use the probability-limit rules for products and inverses used in the lecture's OLS-consistency argument; you do not need to state or prove the named theorems behind these rules.

- (c) Explain why $\mathbb{E}(\varepsilon_i | x_i) = 0$ implies $\mathbb{E}(x_i \varepsilon_i) = 0$. Why is the latter condition weaker?
- (d) Let

$$Q = \begin{pmatrix} 1 & 1/2 \\ 1/2 & 2 \end{pmatrix}, \quad g = \begin{pmatrix} 0.1 \\ 0.4 \end{pmatrix}.$$

Calculate $Q^{-1}g$ and describe where the intercept and slope estimates converge relative to their targets.

- (e) Does the algebraic decomposition change for time-series data? Which assumptions used to justify the probability limits must change?

Solution.

- (a) In stacked form, $y = X\beta + \varepsilon$. Therefore

$$\begin{aligned}\widehat{\beta} &= (X'X)^{-1}X'(X\beta + \varepsilon) \\ &= \beta + (X'X)^{-1}X'\varepsilon,\end{aligned}$$

and $\widehat{\beta} - \beta = (X'X)^{-1}X'\varepsilon$. Conditional on X , the matrix multiplying ε is fixed, so

$$\mathbb{E}(\widehat{\beta} \mid X) = \beta + (X'X)^{-1}X'\mathbb{E}(\varepsilon \mid X) = \beta.$$

Iterated expectations then give $\mathbb{E}(\widehat{\beta}) = \mathbb{E}[\mathbb{E}(\widehat{\beta} \mid X)] = \beta$.

- (b) Multiplying the numerator and denominator by $1/n$ gives

$$\widehat{\beta} - \beta = \left(\frac{1}{n} \sum_{i=1}^n x_i x_i' \right)^{-1} \left(\frac{1}{n} \sum_{i=1}^n x_i \varepsilon_i \right).$$

Using the permitted probability-limit rules for the inverse and product gives

$$\text{plim } \widehat{\beta} = \beta + Q^{-1}g.$$

Thus $g = 0$, together with nonsingular Q and the required LLNs, gives consistency for β .

- (c) By iterated expectations,

$$\mathbb{E}(x_i \varepsilon_i) = \mathbb{E}[x_i \mathbb{E}(\varepsilon_i \mid x_i)] = 0.$$

A zero unconditional cross-moment permits positive and negative conditional relationships to cancel; it does not require the entire conditional mean function to be zero. It is therefore weaker.

- (d) The determinant of Q is $2 - 1/4 = 7/4$, so

$$Q^{-1} = \begin{pmatrix} 8/7 & -2/7 \\ -2/7 & 4/7 \end{pmatrix}.$$

Consequently,

$$Q^{-1}g = \begin{pmatrix} (0.8 - 0.8)/7 \\ (-0.2 + 1.6)/7 \end{pmatrix} = \begin{pmatrix} 0 \\ 0.2 \end{pmatrix}.$$

The intercept estimate converges to its target, while the slope estimate converges to its target plus 0.2. These are statements about probability limits; no convergence of the estimators' expectations has been assumed.

- (e) The identity $\widehat{\beta} - \beta = (X'X)^{-1}X'\varepsilon$ is algebra. It does not depend on whether the observations are cross-sectional or time-series. What changes is the probabilistic justification: an iid LLN is no longer appropriate. One needs time-series conditions such as stability or stationarity, sufficiently weak dependence, finite moments, a nonsingular limiting Q , and an exogeneity condition that makes the limiting numerator zero.

4. This question uses the central limit theorem for approximation and inference.

- (a) Let X_i be iid exponential random variables with rate 1, so that $\mathbb{E}(X_i) = \text{Var}(X_i) = 1$. For $n = 100$, approximate $\Pr(0.8 < \bar{X}_n < 1.15)$. You may use $\Phi(1.5) = 0.9332$ and $\Phi(-2) = 0.0228$. Why can a normal approximation be useful even though an individual exponential draw is strongly right-skewed?

- (b) In a scalar regression without an intercept, assume iid observations, A1, A2, A3' and the finite-moment conditions for Lecture 5's OLS normality result. Suppose

$$Q = \mathbb{E}(x_i^2) = 2, \quad S = \mathbb{E}(x_i^2 \varepsilon_i^2) = 18.$$

Use the OLS asymptotic variance formula $V = Q^{-1}SQ^{-1}$ to find V . For $n = 200$ and $\hat{\beta} = 1.30$, calculate the asymptotic standard error, a 95% confidence interval, the two-sided statistic for $H_0 : \beta = 1$, and its approximate two-sided p -value. You may use $z_{0.975} = 1.96$ and $\Phi(2) = 0.97725$.

- (c) Explain why S , rather than the homoscedastic simplification $\sigma^2 Q$, is the appropriate object in part (b).

Solution.

- (a) The central limit theorem gives

$$\frac{\sqrt{100}(\bar{X}_{100} - 1)}{1} \sim N(0, 1).$$

Hence

$$\begin{aligned} \Pr(0.8 < \bar{X}_{100} < 1.15) &\simeq \Pr\left(\frac{0.8 - 1}{0.1} < Z < \frac{1.15 - 1}{0.1}\right) \\ &= \Pr(-2 < Z < 1.5) = 0.9332 - 0.0228 = 0.9104. \end{aligned}$$

The CLT concerns the standardized sum or average, not the shape of one draw. Averaging many independent finite-variance draws makes the standardized sampling distribution approach a normal shape even when the original density is skewed.

- (b) Since Q is scalar,

$$V = \frac{1}{2}(18)\frac{1}{2} = 4.5.$$

The large-sample approximation to $\text{Var}(\hat{\beta})$ is V/n , so

$$\widehat{se}(\hat{\beta}) \simeq \sqrt{\frac{4.5}{200}} = \sqrt{0.0225} = 0.15.$$

The 95% interval is

$$1.30 \pm 1.96(0.15) = 1.30 \pm 0.294 = [1.006, 1.594].$$

The two-sided test uses the statistic $(1.30 - 1)/0.15 = 2$. Its approximate two-sided p -value is $2[1 - \Phi(2)] = 2(0.02275) = 0.0455$.

- (c) Under the iid and A3' assumptions in part (b), the CLT variance of $n^{-1/2} \sum x_i \varepsilon_i$ is $S = \mathbb{E}(x_i^2 \varepsilon_i^2)$. The equality $S = \sigma^2 Q$ needs conditional homoscedasticity, $\mathbb{E}(\varepsilon_i^2 | x_i) = \sigma^2$. Since no such restriction is given, using S permits conditional heteroscedasticity and is the appropriate robust formula.

5. Suppose

$$Z_n = \sqrt{n}(\hat{\beta}_n - \beta) \xrightarrow{d} Z,$$

where Z has a distribution on the real line that is not concentrated at a single value. You may use the following probability bound: for every $\eta > 0$, there are finite $M > 0$ and N such that $\Pr(|Z_n| > M) < \eta$ for all $n \geq N$. In words, the scaled error stays within a fixed range with arbitrarily high probability once n is large enough.

- (a) Use the supplied bound and the definition of consistency to prove that $\hat{\beta}_n \xrightarrow{p} \beta$.

- (b) Explain the difference between the two conclusions: consistency concerns the unscaled error, while asymptotic normality (or another distributional limit) concerns the shape of a scaled error.
- (c) For these comparisons, use the working approximation that a typical error or standard error has the form $c/\sqrt{n}$. What happens when the sample size is multiplied by 4? By 9? By what factor must the sample size increase to reduce the error or standard error by 40%?

Solution.

- (a) Fix $\varepsilon > 0$ and $\eta > 0$. The supplied probability bound gives finite M and N_1 such that $\Pr(|Z_n| > M) < \eta$ for all $n \geq N_1$. Choose N_2 such that $\sqrt{n}\varepsilon > M$ whenever $n \geq N_2$. Then, for $n \geq \max\{N_1, N_2\}$,

$$\begin{aligned}\Pr(|\hat{\beta}_n - \beta| > \varepsilon) &= \Pr(|Z_n| > \sqrt{n}\varepsilon) \\ &\leq \Pr(|Z_n| > M) < \eta.\end{aligned}$$

Since η was arbitrary, the probability tends to zero; hence $\hat{\beta}_n \xrightarrow{p} \beta$.

- (b) Consistency says that the unscaled distribution collapses onto the fixed target. A limit distribution for the scaled error that does not collapse to a point describes the remaining local fluctuations after magnifying them. It supplies a rate and a limiting shape, whereas consistency alone supplies neither.
- (c) Under the working approximation, the error or standard error is $c/\sqrt{n}$. Multiplying n by 4 therefore approximately halves it, and multiplying n by 9 approximately divides it by 3. A 40% reduction leaves 60% of the original value. Thus, if m is the sample-size multiplier,

$$\frac{1}{\sqrt{m}} = 0.6 \quad \implies \quad m = \frac{1}{0.6^2} = 2.777\dots$$

The approximation therefore requires about 2.78 times as many observations.

Non-examinable extension (for self-learning only)

6. Let $\beta > 0$ and suppose $\sqrt{n}(\hat{\beta}_n - \beta) \xrightarrow{d} N(0, V)$, with $V > 0$. Assume the estimates $\hat{\beta}_n$ and $\hat{V}$ used below are positive.
- (a) The delta method states that the asymptotic variance after a differentiable transformation h is $[h'(\beta)]^2 V$. Apply it to $h(\beta) = \log \beta$.
 - (b) If $\hat{V} \xrightarrow{p} V$ and $\hat{\beta}_n \xrightarrow{p} \beta$, give a consistent estimated standard error for $\log \hat{\beta}_n$ and a test statistic obtained by dividing the estimation error by that estimated standard error. Use Slutsky's theorem to state its limit.

Solution.

- (a) Since $h'(\beta) = 1/\beta$, the delta method gives

$$\sqrt{n} [\log \hat{\beta}_n - \log \beta] \xrightarrow{d} N\left(0, \frac{V}{\beta^2}\right).$$

- (b) A consistent estimated standard error is

$$\widehat{se}(\log \hat{\beta}_n) = \sqrt{\frac{\hat{V}}{n\hat{\beta}_n^2}} = \frac{\sqrt{\hat{V}}}{\sqrt{n}|\hat{\beta}_n|}.$$

Because $\beta > 0$ and $\hat{\beta}_n \xrightarrow{p} \beta$, the estimate is positive with probability approaching one. Slutsky's theorem gives the test statistic's limit

$$\frac{\log \hat{\beta}_n - \log \beta}{\sqrt{\hat{V}/(n\hat{\beta}_n^2)}} = \frac{\sqrt{n}|\hat{\beta}_n|(\log \hat{\beta}_n - \log \beta)}{\sqrt{\hat{V}}} \xrightarrow{d} N(0, 1).$$

The unknown target appears only as the centering value, as in Lecture 4's test statistics; the scale is estimated from the data.